A REPRESENTABLE FLAT, LOCALLY FINITELY PRESENTED SURJECTION IS SURJECTIVE ON SHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Definition of injective and surjective maps of sheaves. Sites, Definition sites-definition-sheaves-injective-surjective, used at proof lines 912-914. Is the notion of surjectivity that is proved.
- Representability of the transformation a : F -> G. used at proof lines 907-909. Makes T' = F x_G T a scheme.

Source reference. Original definition 00WM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1188–1188. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $F, G : (Sch/S)^{opp}_{fppf} \to Sets$ be sheaves. Let $a : F \to G$ be representable, flat, locally of finite presentation, and surjective. Then $a : F \to G$ is surjective as a map of sheaves.*

Proof. Let T be a scheme over S and let $g : T \to G$ be a T-valued point of G. By assumption $T' = F \times_G T$ is (representable by) a scheme and the morphism $T' \to T$ is a flat, locally of finite presentation, and surjective. Hence $\{T' \to T\}$ is an fppf covering such that $g|_{T'} \in G(T')$ comes from an element of $F(T')$, namely the map $T' \to F$. This proves the map is surjective as a map of sheaves, see Sites, Definition 00WM. □

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.